Supporting Information

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SI Materials and Methods

Isolation and Preparation of DNA for Whole-Genome Replication Profile Analysis. ***EdU incorporation of synchronous yeast cells in hydroxyurea.*** Yeast cells were synchronized in G1 with α-factor and were released into yeast extract/peptone/dextrose (YPD) medium containing 0.2 mg/mL pronase E, 0.2 M hydroxyurea (HU), and 0.5 mM 5-ethynyl-2'-deoxyuridine (EdU). At the indicated time point, NaN_3 was added to a final concentration of 0.1% to stop all cellular functions, and cells were collected by centrifugation. About 5×10^8 cells were used for preparation of each genomic DNA sample.

Preparation of genomic DNA. About 5×10^8 cells labeled with EdU were washed with 20 mL of buffer containing 10 mM Tris·HCl (pH 7.5) and 50 mM EDTA. Cells then were resuspended in 0.8 mL of nuclear isolation buffer [50 mM 3-(*N*-morpholino)propanesulfonic acid, 150 mM potassium acetate, 17% glycerol, 2 mM magnesium chloride, 500 μM spermidine, and 150 μM spermine, adjusted to pH 7.2 with KOH]. An equal (cell) volume of zirconium beads was added to the cell suspension, and cells were lysed at 4 °C by bead beating using a Baxter Multi-Tube Vortexer, vortexing for 17–20 times with a 1 min-on/1 min-off cycle. Cell breakage was verified by microscopy, and ∼90% of cells should become ghost. Suspensions with broken cells were pelleted by centrifugation at $6{,}500 \times g$ at 4 °C for 10 min. The pellets were resuspended in 1.8 mL of Qiagen Buffer G2 containing 200 μg/mL RNase A and then were incubated at 37 °C for 30 min. Proteinase K was added to a final concentration of 0.5 mg/mL, and the suspension was incubated for 1 h further at 37 °C. The supernatant was collected after centrifugation at $6{,}500 \times g$ for 10 min at 4 °C. An equal volume of Qiagen Buffer QBT was added to each sample, and genomic DNA was isolated using a Qiagen Genomic-tip 20/G column (catalog no. 10223) according to the manufacturer's instructions. The genomic DNA preparation was resuspended in 40 μL of EB buffer (10 mM Tris·Cl, pH 8.5) (Qiagen).

Preparation and purification of EdU-genomic DNA for high-throughput DNA sequencing. About 1.2 μg of each genomic DNA sample was used as starting material for preparing libraries for high-throughput sequencing. Each genomic DNA sample was resuspended in 80 μL of EB buffer and was fragmented by sonication using Bioruptor to a size range of ∼100 to ∼550 bp. Quality of fragmentation was verified on 2% agarose gel using 5 μL of the sonicated DNA sample. The remaining 75 μL was subjected to end repair, A tailing, and adaptor ligation using reagents from New England Biolabs and the protocol described in the NEBNext DNA library preparation instruction manual. The adaptor oligos containing custom barcodes (Table S2) were used to prepare these DNA libraries. EdU-labeled genomic DNA was biotinylated using reagents and the Click reaction protocol described in the Invitrogen RNA Capture kit (C10365). The biotinylated EdU-genomic DNA samples then were purified using Streptavidin T1 magnetic beads (Invitrogen). For each pull-down, 20 μL of magnetic beads and 30 μL of DNA sample were incubated with an additional 150 μL of buffer containing 10 mM Tris·HCl (pH 7.5), 1 M NaCl, 10 mM EDTA, 0.02% Nonidet P-40, and 1 mg/mL BSA at 25 °C for 30 min, then were washed five times (5 min each wash) with buffer containing 10 mM Tris·HCl (pH 7.5), 1 M NaCl, 10 mM EDTA, and 0.02% Nonidet P-40, and were eluted twice at 55 °C for 1 h with 100 μL of elution buffer containing 10 mM Tris·HCl (pH 8), 10 mM EDTA, and 0.1% SDS. The two sequential eluates were pooled and purified using MinElute column (Qiagen) and were eluted in 25 μL EB buffer. The purified, barcoded, biotinylated DNA was PCR-amplified using primers SR 1.1 and SR 2.1 (Table S2) for 14 cycles and then was quantified, pooled and submitted for sequencing according to the guidelines of the sequencing facility.

Computational Analyses of Sequencing Data. ***Read mapping.*** Using the FASTX toolkit (http://hannonlab.cshl.edu/fastx_toolkit/index.html), single-end reads (36 or 101 bp) were filtered for quality (90% of positions were required to have a quality score ≥10) and organized according to the 5-bp barcode on the 5′ end of each read, which then was removed. Trimmed reads were mapped to the *Saccharomyces cerevisiae* S288C genome (S288C reference genome version R64-1-1_20110203 downloaded from http://yeastgenome.org on February 18, 2011) using Burrows–Wheeler aligner (1) with default parameters. Pileup files were generated using SAMtools (2), and the number of reads covering every 31st nucleotide position in the genome was recorded using a custom Python script. Positions in the genome with unreliable mapping were determined as follows. Every overlapping 31-bp interval of the S288C genome first was mapped back to the genome; then the number of these simulated reads covering each 31st nucleotide position was determined as above. Genomic positions at which the number of overlapping simulated reads differed from the number expected were flagged.

Replication profiles. Read counts at 31-bp intervals were averaged across the genome using a sliding window of 2,000 bp (all other samples), yielding a replication profile. These window sizes were selected manually to provide reasonable smoothness for the resulting peaks while ensuring that window size remained substantially smaller than mean peak width. Each replication profile was computed twice, first using a sliding window that excluded flagged positions and then using one that did not. Positions at which the former profile was less than 80% of the latter then were masked, as indicated by gaps in the genomic and chromosomal replication profiles shown. Because of the nature of the data, the absolute scale of each replication profile (i.e., the height corresponding to fully replicated DNA) could not be determined unambiguously. Instead, we recorded the number of reads bounding 99.5% of positions within each profile. The corresponding profile then was divided by this read count to yield a normalized profile. This normalization procedure is an important consideration in interpreting figures showing overlaid profiles from multiple samples.

Peak width. Peak widths observed in replication profiles were computed for annotated autonomously replicating sequences (ARSs) as follows. For each ARS annotated in OriDB (3), we examined the region of the genome lying closer to the center of this ARS than to the center of any other. The maximum profile height within this region was recorded as peak height. A peak width was recorded if, on each side of this maximum and within the stated interval, the replication profile fell below half maximum without crossing any masked genomic positions. In this case peak width was defined as the full width of the peak at half maximum. Only ARSs having a well-defined peak width are included in plots of width vs. height. Width-vs.-width plots show only ARSs having defined peak widths and normalized peak heights greater than 0.1 in both samples; this minimum height is indicated by a dashed horizontal line in width-vs.-height plots.

1. Li H, Durbin R (2009) Fast and accurate short read alignment with Burrows-Wheeler transform. *Bioinformatics* 25(14):1754–1760.
2. Li H, et al.; 1000 Genome Project Data Processing Subgroup (2009) The Sequence Alignment/Map format and SAMtools. *Bioinformatics* 25(16):2078–2079.
3. Nieduszynski CA, Hiraga S, Ak P, Benham CJ, Donaldson AD (2007) OriDB: A DNA replication origin database. *Nucleic Acids Res* 35(Database issue):D40–D46.

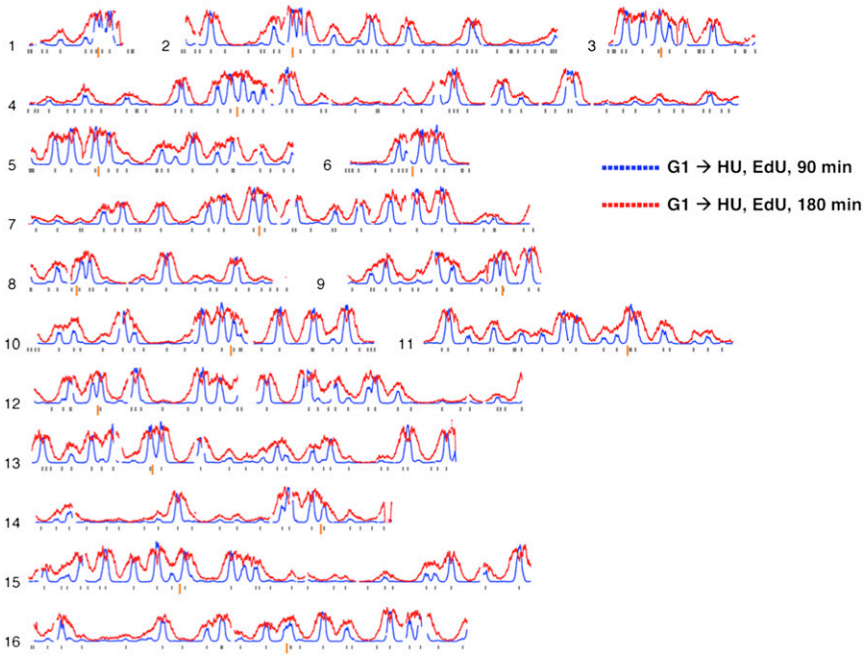

Fig. S1. Whole-genome replication profile analysis. YS2571 wild-type cells were synchronized in G1 phase and released into YPD medium containing 0.2 M HU and 0.5 mM EdU for 90 min (blue) or 180 min (red). EdU-labeled DNA was isolated and sequenced. Each line shows a DNA replication profile for each chromosome from cells replicating in HU. Gray bars under the profiles indicate the annotated replication origins according to OriDB (3). Orange bars indicate centromeres.

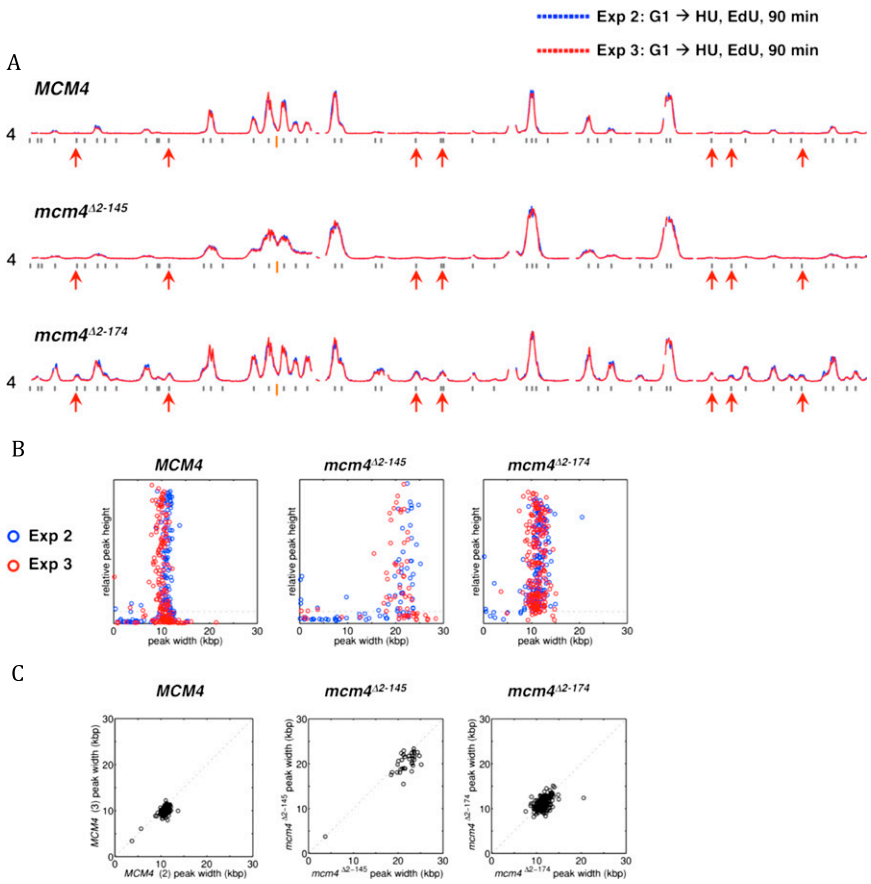

Fig. S2. Reproducibility of DNA replication profile and analysis. (*A*) Replication profiles of chromosome IV for the wild-type cells (YS2571) and *mcm4* mutants (YS2574, YS2585, and YS2758). Red arrows indicate late origins that are not active in HU in wild-type cells. (*B*) Peak width vs. peak height analysis. Data for the same strain from two different experiments are plotted in red and blue. The peak width and peak height for each separate origin is represented as a circle. (*C*) Peak width vs. peak width analysis for each origin from various strains. Data from the same strains obtained in two independent experiments (*x* axis: experiment 2; *y* axis: experiment 3) were compared and plotted to determine the reproducibility the measurement of the replication fork progression.

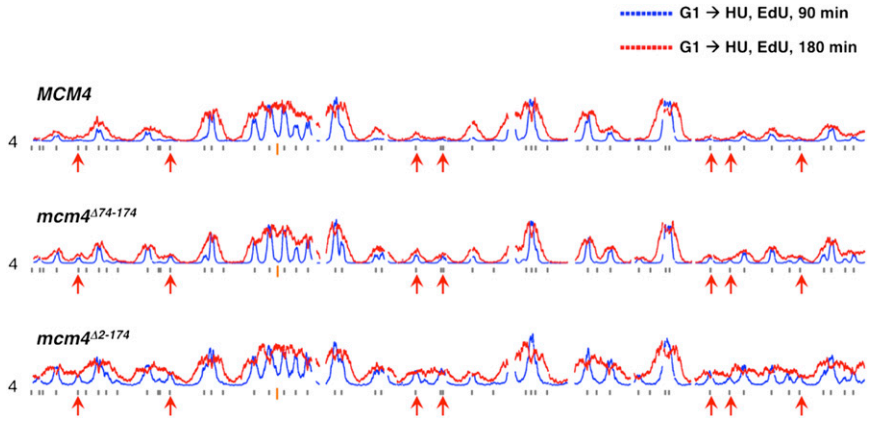

Fig. S3. Progression of the replication fork from 90–180 min in HU. Replication profiles of chromosome IV for the wild-type cells and *mcm4* mutants with truncated N-terminal serine/threonine-rich domain (NSD) segments. Experimental conditions and symbols are as in Fig. S1. Red arrows indicate origins that are suppressed in the wild-type cells but fire in the mutants lacking the proximal NSD.

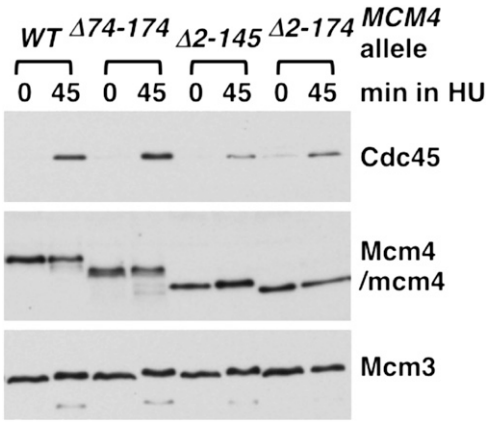

Fig. S4. Analysis of chromatin-bound proteins. Cells synchronized in G1 were released into 0.2 M HU and collected at the indicated time points. Chromatin-bound proteins were extracted from cells and analyzed by Western blot.

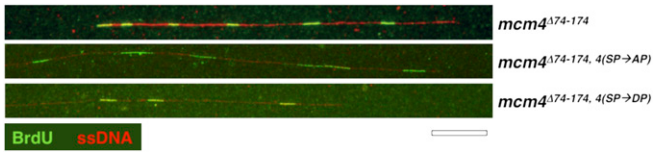

Fig. S5. Single-molecule DNA fiber analysis. Representative images of DNA fiber analyzed from *mcm4*$^{\Delta 74–174}$, *mcm4*$^{\Delta 74–174,\ 4(SP\rightarrow AP)}$, and *mcm4*$^{\Delta 74–174,\ 4(SP\rightarrow DP)}$ cells. BrdU is labeled with an anti-BrdU antibody (green), and DNA is counterstained with an antibody against ssDNA (red). (Scale bar: 50 kb.) Cells were synchronized in G1 phase and released into S phase with pronase (50 μg/mL) in the presence of 0.2 M HU and 400 μg/mL of BrdU for 90 min.

Table S1. Yeast strains used in this study

Strain	Genotype	Source
YS2571	*MATa bar1Δ::TRP1 URA3::BrdU-Inc ade2-1 can1-100 his3-11,-15 leu2-3,112 trp1-1 ura3-1*	This study
YS2758	*MATa mcm4*$^{\Delta 2–145}$ *bar1Δ::TRP1 URA3::BrdU-Inc ade2-1 can1-100 his3-11,-15 leu2-3,112 trp1-1 ura3-1*	This study
YS2585	*MATa mcm4*$^{\Delta 2–174}$ *bar1Δ::TRP1 URA3::BrdU-Inc ade2-1 can1-100 his3-11,-15 leu2-3,112 trp1-1 ura3-1*	This study
YS2574	*MATa mcm4*$^{\Delta 74–174}$ *bar1Δ::TRP1 URA3::BrdU-Inc ade2-1 can1-100 his3-11,-15 leu2-3,112 trp1-1 ura3-1*	This study
YS2755	*MATa mcm4*$^{5(SP\rightarrow AP)}$ *bar1Δ::TRP1 URA3::BrdU-Inc ade2-1 can1-100 his3-11,-15 leu2-3,112 trp1-1 ura3-1*	This study
YS2813	*MATa mcm4*$^{5(SP\rightarrow DP)}$ *bar1Δ::TRP1 URA3::BrdU-Inc ade2-1 can1-100 his3-11,-15 leu2-3,112 trp1-1 ura3-1*	This study
YS2761	*MATa mcm4*$^{\Delta 74–174,\ 4(SP\rightarrow AP)}$ *bar1Δ::TRP1 URA3::BrdU-Inc ade2-1 can1-100 his3-11,-15 leu2-3,112 trp1-1 ura3-1*	This study
YS2764	*MATa mcm4*$^{\Delta 74–174,\ 4(SP\rightarrow DP)}$ *bar1Δ::TRP1 URA3::BrdU-Inc ade2-1 can1-100 his3-11,-15 leu2-3,112 trp1-1 ura3-1*	This study
YS2251	*MATa bar1Δ::TRP1 ade2-1 can1-100 his3-11,-15 leu2-3,112 trp1-1 ura3-1*	(1)
YS2263	*MATa mcm2*$^{1-200}$*-mcm4*$^{\Delta 2–174}$ *bar1Δ::TRP1 ade2-1 ura3-1 his3-11,15 trp1-1 leu2-3,112 can1-100*	(1)
YS2534	*MATa mcm4*$^{\Delta 2–174}$ *bar1Δ::TRP1 ade2-1 can1-100 his3-11,-15 leu2-3,112 trp1-1 ura3-1*	This study
YS2754	*MATa mcm4*$^{\Delta 38–174}$ *bar1Δ::TRP1 ade2-1 can1-100 his3-11,-15 leu2-3,112 trp1-1 ura3-1*	This study
YS2261	*MATa mcm4*$^{\Delta 74–174}$ *bar1Δ::TRP1 ade2-1 ura3-1 his3-11,15 trp1-1 leu2-3,112 can1-100*	(1)

All strains used in this study are of W303 background.

1. Sheu YJ, Stillman B (2010) The Dbf4-Cdc7 kinase promotes S phase by alleviating an inhibitory activity in Mcm4. *Nature* 463(7277):113–117.

Table S2. Barcode adaptors and oligos used for library construction

Primer name	Sequence from 5′ end
SR1.1	AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACGACGCTCTTCCGATCT
SR2.1	CAAGCAGAAGACGGCATACGAGCTCTTCCGATCT
552W-SRadBC1	ACACTCTTTCCCTACACGACGCTCTTCCGATCTGATC*T
553C-SRadBC1	p-GATCAGATCGGAAGAGCTCGTATGCCGTCTTCTGCTTG
554W-SRadBC2	ACACTCTTTCCCTACACGACGCTCTTCCGATCTATCG*T
555C-SRadBC2	p-CGATAGATCGGAAGAGCTCGTATGCCGTCTTCTGCTTG
556W-SRadBC3	ACACTCTTTCCCTACACGACGCTCTTCCGATCTTGTC*T
557C-SRadBC3	p-GACAAGATCGGAAGAGCTCGTATGCCGTCTTCTGCTTG
558W-SRadBC4	ACACTCTTTCCCTACACGACGCTCTTCCGATCTGTGA*T
559C-SRadBC4	p-TCACAGATCGGAAGAGCTCGTATGCCGTCTTCTGCTTG
574W-SRadBC5	ACACTCTTTCCCTACACGACGCTCTTCCGATCTAGAG*T
575C-SRadBC5	p-CTCTAGATCGGAAGAGCTCGTATGCCGTCTTCTGCTTG
576W-SRadBC6	ACACTCTTTCCCTACACGACGCTCTTCCGATCTCATA*T
577C-SRadBC6	p-TATGAGATCGGAAGAGCTCGTATGCCGTCTTCTGCTTG
578W-SRadBC7	ACACTCTTTCCCTACACGACGCTCTTCCGATCTTGCG*T
579C-SRadBC7	p-CGCAAGATCGGAAGAGCTCGTATGCCGTCTTCTGCTTG
580W-SRadBC8	ACACTCTTTCCCTACACGACGCTCTTCCGATCTGCAC*T
581C-SRadBC8	p-GTGCAGATCGGAAGAGCTCGTATGCCGTCTTCTGCTTG
582W-SRadBC9	ACACTCTTTCCCTACACGACGCTCTTCCGATCTCTGG*T
583C-SRadBC9	p-CCAGAGATCGGAAGAGCTCGTATGCCGTCTTCTGCTTG
584W-SRadBC10	ACACTCTTTCCCTACACGACGCTCTTCCGATCTACTG*T
585C-SRadBC10	p-CAGTAGATCGGAAGAGCTCGTATGCCGTCTTCTGCTTG
586W-SRadBC11	ACACTCTTTCCCTACACGACGCTCTTCCGATCTTAGC*T
587C-SRadBC11	p-GCTAAGATCGGAAGAGCTCGTATGCCGTCTTCTGCTTG
588W-SRadBC12	ACACTCTTTCCCTACACGACGCTCTTCCGATCTCACG*T
589C-SRadBC12	p-CGTGAGATCGGAAGAGCTCGTATGCCGTCTTCTGCTTG
590W-SRadBC13	ACACTCTTTCCCTACACGACGCTCTTCCGATCTGCTA*T
591C-SRadBC13	p-TAGCAGATCGGAAGAGCTCGTATGCCGTCTTCTGCTTG
592W-SRadBC14	ACACTCTTTCCCTACACGACGCTCTTCCGATCTATAC*T
593C-SRadBC14	p-GTATAGATCGGAAGAGCTCGTATGCCGTCTTCTGCTTG

p-, 5′ phosphate.
*Phosphorothioate modification.